

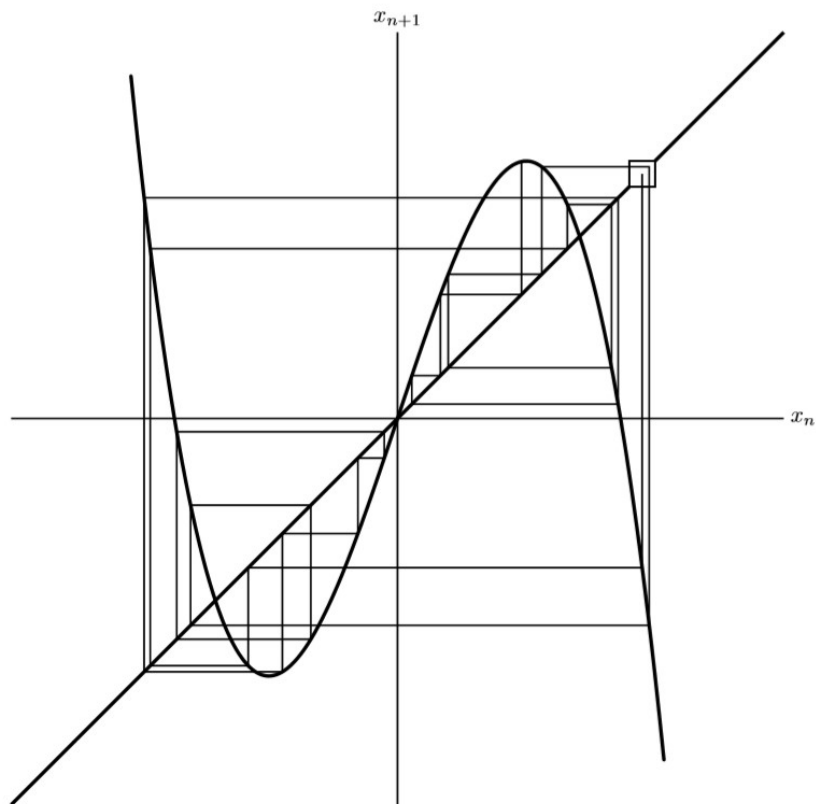
10.1.11

a)

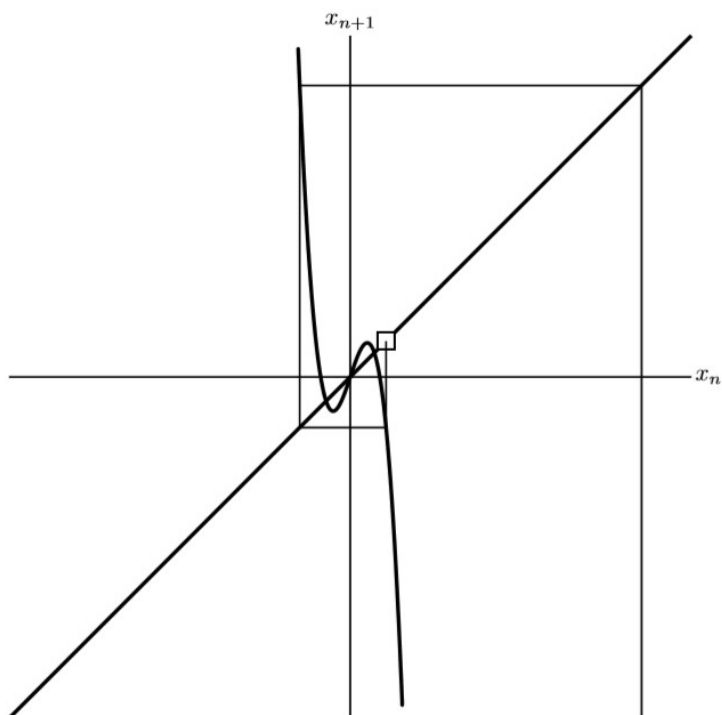
$$x = 3x - x^3 \Rightarrow 0 = 2x - x^3 = x(x + \sqrt{2})(x - \sqrt{2})$$

So the fixed points are $x = 0, \pm\sqrt{2}$.

b)



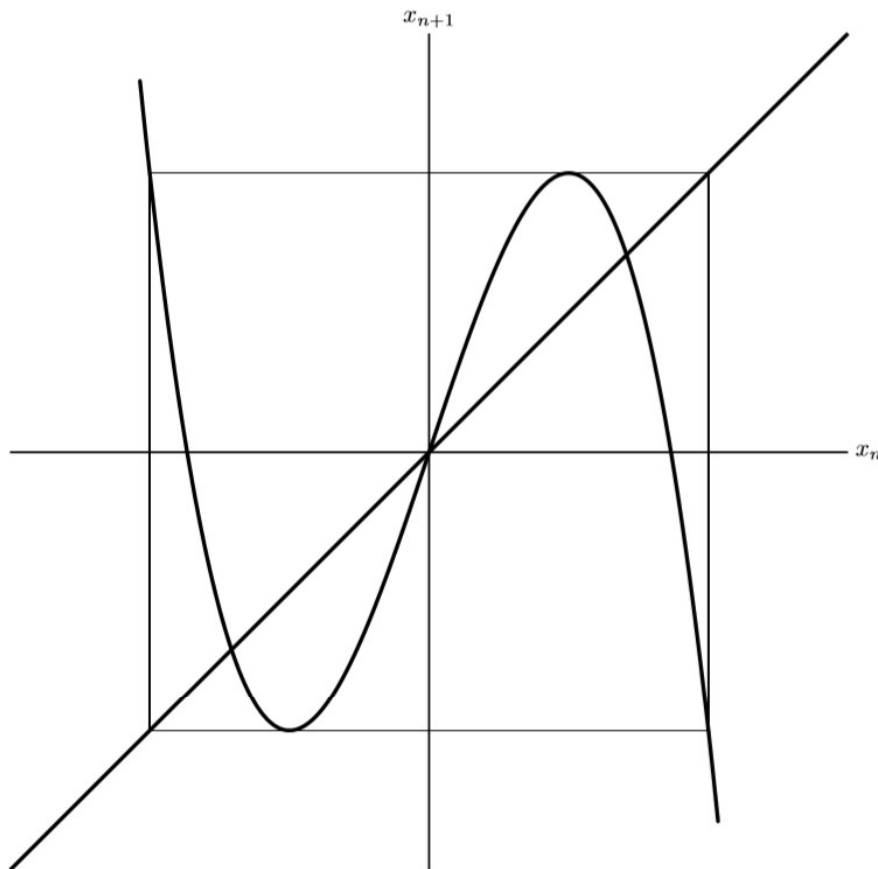
c)



10.1.11 (cont.)

d)

There exists a 2-cycle, $x_n = \pm 2 \Rightarrow x_{n+1} = 3(\pm 2) - (\pm 2)^3 = \mp 2$ and $2 \rightarrow -2 \rightarrow 2 \rightarrow \dots$ which is the reason for the change in behavior from $x_0 = 1.9$ to $x_0 = 2.1$.



If $-2 < x_n < 2$ then $-2 < x_{n+1} = 3x_n - x_n^3 < 2$ simply because the graph fits in the box made by the 2-cycle and thus every point maps to a point in the box.

As for starting outside the 2-cycle box

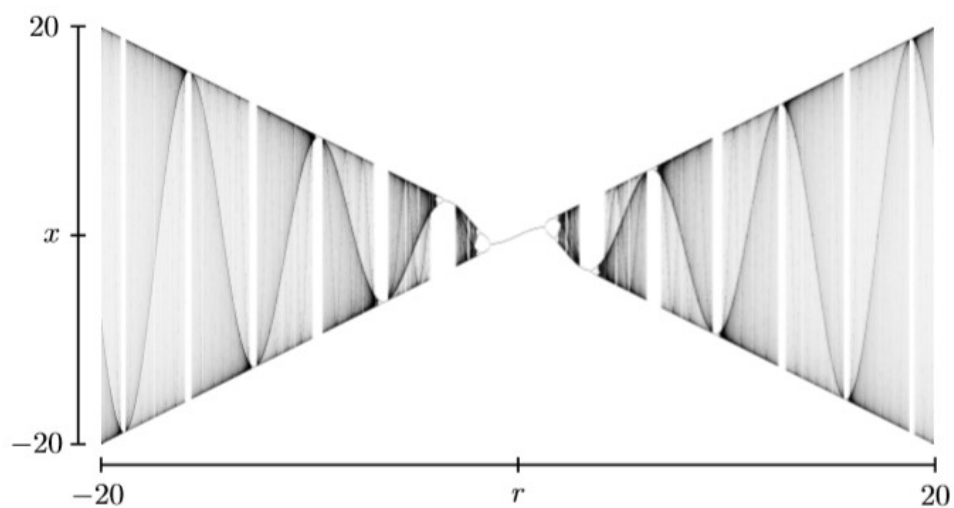
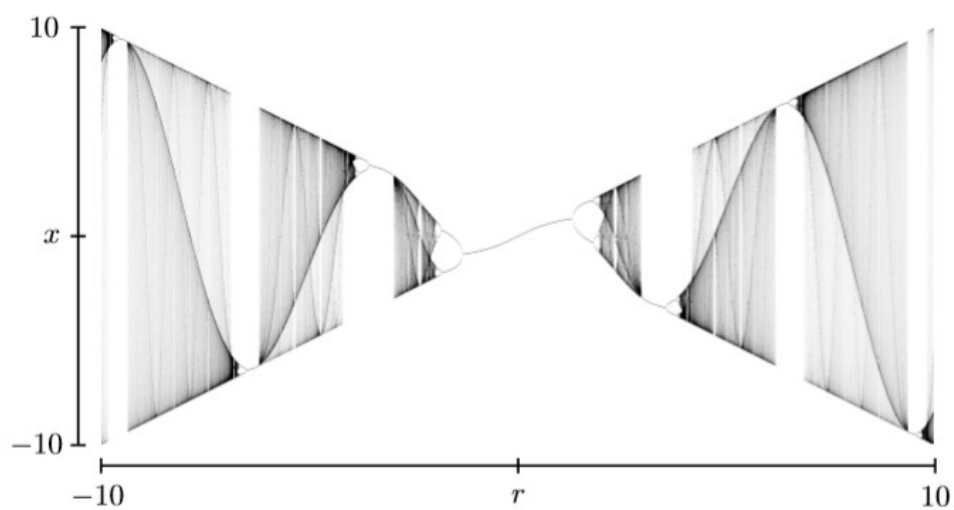
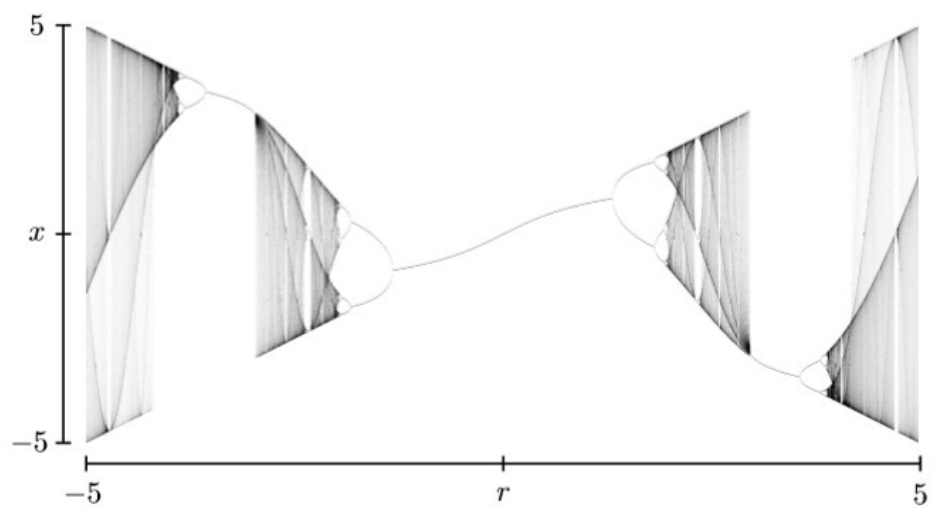
$$|x_n| = 2 + \epsilon \quad \epsilon > 0$$

$$\begin{aligned} |x_1| &= |3x_0 - x_0^3| = |x_0||3 - x_0^2| = |x_0||3 - (2 + \epsilon)^2| \\ &= |x_0||3 - (4 + 2\epsilon + \epsilon^2)| = |x_0||-1 - 2\epsilon - \epsilon^2| \\ &= |x_0|(1 + 2\epsilon + \epsilon^2) \end{aligned}$$

$$|x_n| = |x_0|(1 + 2\epsilon + \epsilon^2)^n \Rightarrow \lim_{n \rightarrow \infty} |x_n| = \infty$$

So the iteration map will always be increasing in magnitude and spiraling away from the 2-cycle.

10.2.6



10.3.1

$$x = f(x) = rx(1 - x) = rx - rx^2$$

$$rx^2 + (1 - r)x = x(rx + 1 - r) = 0 \Rightarrow x^* = 0, \frac{r - 1}{r}$$

$$f'(x) = r(1 - 2x)$$

$$f'(0) = r = 0$$

$x^* = 0$ is a superstable fixed point when $r = 0$

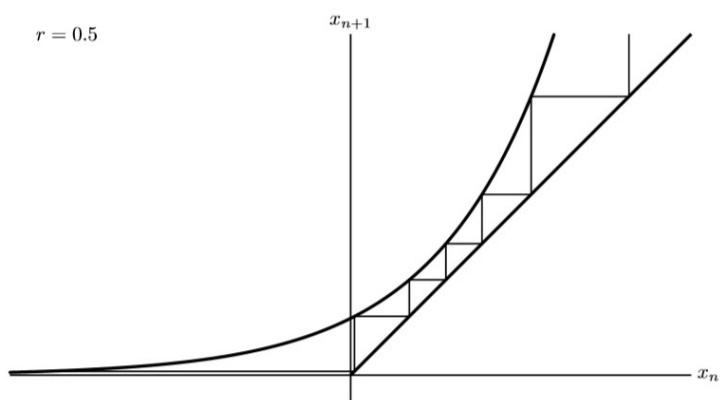
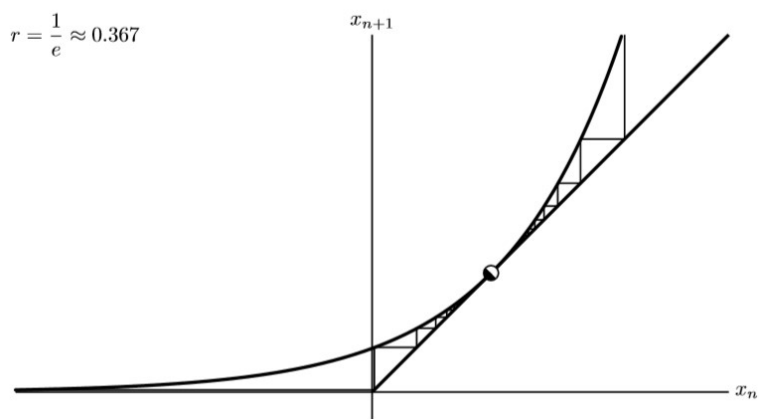
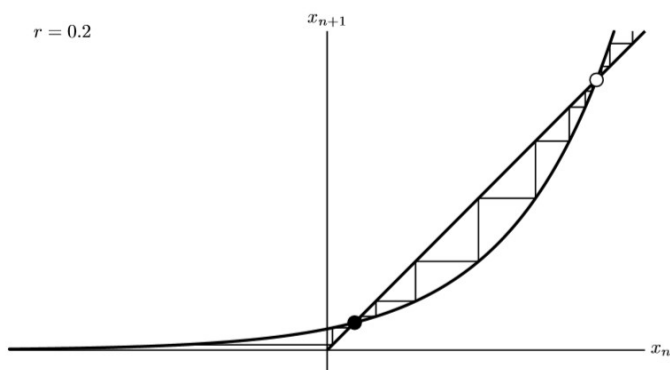
$$f'\left(\frac{r - 1}{r}\right) = r\left(1 - 2\frac{r - 1}{r}\right) = 0 \Rightarrow r = 0, 2$$

$x^* = \frac{r - 1}{r}$ is a superstable fixed point when $r = 2$

$x^* = 0$ is superstable if $r = 0$ because $x_1 = 0$ no matter what x_0 is. The other fixed point is superstable when $r = 2$ but not when $r = 0$ because $x_1 = 0$ as before.

10.4.1

a)



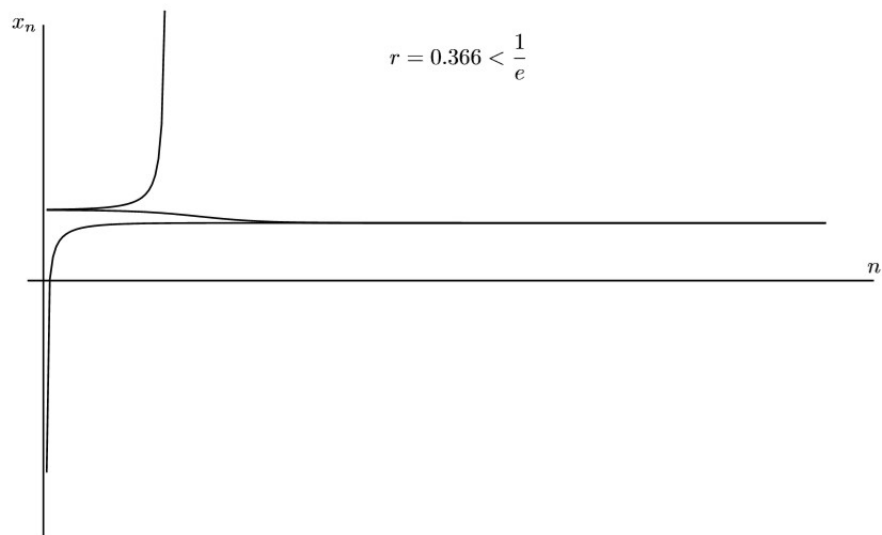
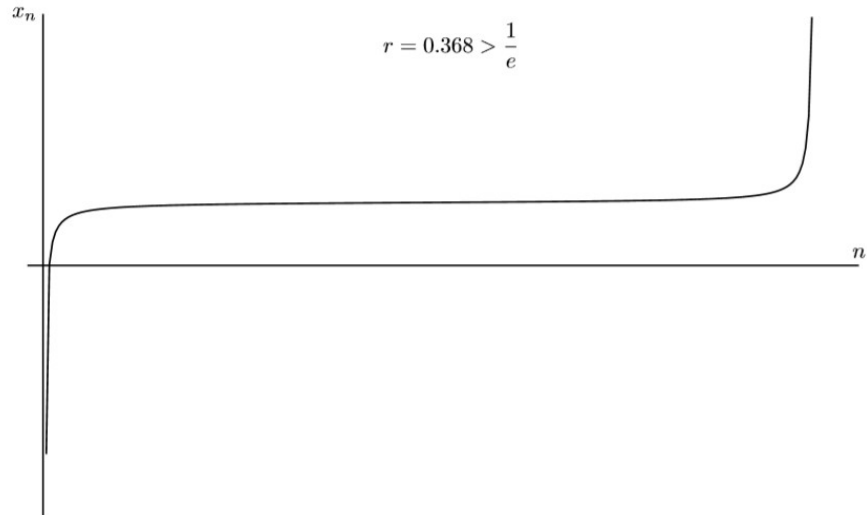
10.4.1 (cont.)

b)

A tangent bifurcation does occur at $r = \frac{1}{e}$ as is evident in the graphs. We can verify by confirming that the graph of the iteration function lies tangent to the diagonal line.

$$r = \frac{1}{e} \quad x = 1 \Rightarrow re^x = x \quad \frac{d}{dx} re^x = 1$$

c)



Here you can see the ghost of the tangent bifurcation. The first graph takes quite a while for the trajectory to iterate through, but the second graph still has the unstable and stable fixed points in existence.

Note: The graphs look continuous, but that really is an artifact of some many points near the ghost of the tangent bifurcation and connecting the dots.